

L1.2 Limits

§1.5–1.6 — what you need to know

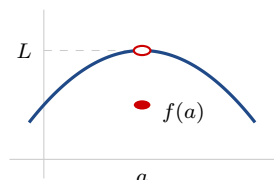
WHAT THE LIMIT IS

$$\lim_{x \rightarrow a} f(x) = L \iff \text{LHL} = \text{RHL} = L$$

“read the neighborhood, ignore the dot”

$f(a)$ can be missing, or present and wrong, and the limit does not move.

Notation. LHL RHL DNE ($\sim \exists, \nexists$) · H $\emptyset$ hole · Z $\emptyset$ zero · VA $\emptyset$ · HA $\emptyset$



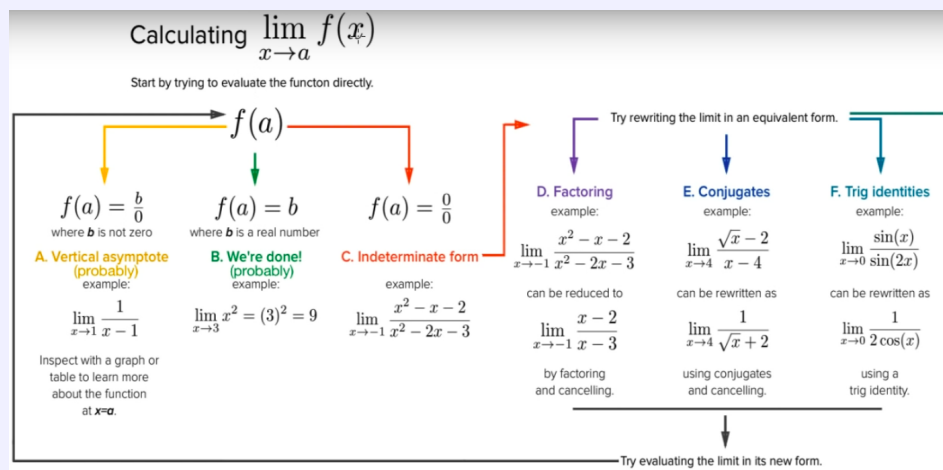
$f(a)$ IS A DIAGNOSTIC

$$\lim_{x \rightarrow a} f(x) = f(a) \text{ if } f \text{ is polynomial, rational or trig, and } a \in D$$

“you don’t evaluate $f(a)$ to get the answer — you evaluate it to find out what situation you’re in”

This hypothesis is what (*probably*) means on the chart, and it runs out exactly one way: f is **piecewise** at a . $[1] = 1$ reads clean; $\lim_{x \rightarrow 1} [x]$ is DNE.

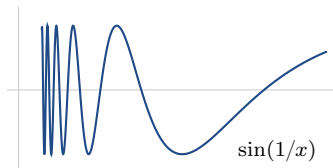
The chart — try $f(a)$, then read what came back



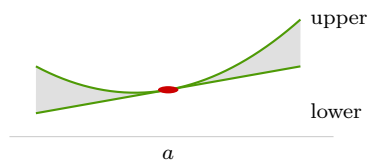
Not on here yet: limits at infinity · L'Hôpital (Unit 2) · the special trig limits (§2.4). You add them here later.

WHEN THE DIAGNOSTIC DOESN'T WORK — TWO TRIGGERS

A blow-up trapped inside sin or cos. $\sin(1/0)$ is not a number and not one of the three readings. **Move: graph it.** Then one question — *does the amplitude collapse?* No $\Rightarrow$ DNE. Yes $\Rightarrow$ the limit is 0.



The problem hands you bounds. There is no f to evaluate — the diagnostic never starts. **Move: squeeze.** Evaluate both bounds at a ; if they agree, that value is the limit. Check they really do sandwich near a .



THE LIMIT LAWS — EVERY LIMIT BELOW IS AS $x \rightarrow a$

One move: **replace each function by its limit and do the arithmetic.** The laws do not tell you how — they tell you when you are allowed to. **1–9 need $\lim f$ and $\lim g$ to both exist.**

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| 1. $\lim[f + g] = \lim f + \lim g$ | 6. $\lim[f]^n = [\lim f]^n$ |
| 2. $\lim[f - g] = \lim f - \lim g$ | 7. $\lim c = c \cdot \lim x = a$ |
| 3. $\lim[cf] = c \lim f$ | 8. $\lim x^n = a^n \cdot \lim \sqrt[n]{x} = \sqrt[n]{a}$ |
| 4. $\lim[fg] = \lim f \cdot \lim g$ | 9. $\lim \sqrt[n]{f} = \sqrt[n]{\lim f}$ |
| 5. $\lim \frac{f}{g} = \frac{\lim f}{\lim g}$ if $\lim g \neq 0$ | 10. $\lim f(g(x)) = f(\lim g)$ if f cont. |

5 and **10** are the whole content: **5** also needs $\lim g \neq 0$; **10** sits outside the box because it also needs f continuous where g lands — its graph cannot jump there.

WATCH OUT

- ✗ it gets closer and closer but never reaches $\rightarrow$ the limit *is* a number, not a journey
- ✗ $\frac{0}{0}$ means the limit is 0, or that there isn't one $\rightarrow$ it means *rewrite and run the diagnostic again*
- ✗ $f(a)$ undefined, so there is no limit $\rightarrow$ the limit ignores $x = a$ entirely
- ✗ a clean number is always the limit $\rightarrow$ not when f is piecewise at a — $[1] = 1$ and the limit is DNE
- ✗ $\frac{b}{0}$, so the answer is ∞ $\rightarrow$ $\frac{b}{0}$ says it blows up, not which way — check both sides
- ✗ squeeze needs a formula for f $\rightarrow$ you are never given one; evaluate the two bounds